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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

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Midterm Examination (MTA) - Makeup
Summer Semester 2025/2026

Time Allowed: 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.



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PART 1: Multiple Choice Questions [10 Marks]

Questions 1 [10 marks] (1 mark/each):

1. Which statement best represents cybersecurity?
   A. It is limited to password creation only
   B. It protects cyberspace using policies, safeguards, technologies, and practices
   C. It is only concerned with physical building security
   D. It is used only to increase computer speed

2. An active attack is an attack that:
   A. Only listens to network traffic
   B. Monitors communication without changing it
   C. Modifies data, disrupts services, or gains unauthorized access
   D. Does not affect system security

3. Information security is mainly concerned with preserving:
   A. Cost, speed, and usability
   B. Confidentiality, integrity, and availability
   C. Marketing, storage, and design
   D. Encryption, compression, and backup only

4. Which of the following is a correct description of asymmetric encryption?
   A. It uses one shared key for all users
   B. It uses a public key and a private key
   C. It produces only a hash value
   D. It does not require any keys

5. A cryptographic hash function is mainly used to produce:
   A. A fixed-length digest from input data
   B. A user role for access control
   C. A decrypted message
   D. A privacy regulation

6. Sensitive PII is information that, if disclosed, could:
   A. Improve system performance
   B. Cause harm, embarrassment, inconvenience, or unfairness
   C. Identify only software applications
   D. Always be public information

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7. De-identification aims to:

A. Remove or mask PII to reduce identification risk
B. Increase the amount of collected PII
C. Publish all personal data online
D. Replace all access controls

8. Repurposing collected data means:

A. Destroying data immediately after collection
B. Using data for a new purpose beyond the original intent
C. Encrypting data before storage
D. Preventing all data analysis

9. CVSS is used to assess vulnerabilities based on:

A. Interface design and usability
B. Legal documents only
C. Exploitability and impact metrics
D. Storage prices

10. The principle of least privilege means that users should be given:

A. All available permissions
B. The same permissions as administrators
C. Only the minimum access required for their tasks
D. Random permissions selected by the system

PART 2: Short Answer Questions [25 Marks]

P2-Q1 [5 marks]: Define Personally Identifiable Information (PII) and provide two examples.

P2-Q2 [5 marks]: Explain the difference between symmetric and asymmetric encryption, including one advantage of each.

P2-Q3 [5 marks]: Explain the purpose of a privacy risk assessment and how it supports organizational decision-making.

P2-Q4 [5 marks]: Discuss how advances in information technology can increase privacy threats.

P2-Q5 [5 marks]: Compare Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC).

PART 3: Problem Solving and Case Studies [25 Marks]

P3-Q1 [20 marks]: Study the following cases and answer the related questions

CASE-1 [3 marks] **"A clinic allows doctors to access patient prescriptions through an online portal. One doctor's password is stolen, and the attacker changes several prescriptions"**. Suggest two security mechanisms to reduce the chance of this attack happening again.

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CASE-2 [3 marks]: **"A new supplier and a company need to exchange a confidential contract online, but they have never shared a secret key before".** Which type of encryption should be used initially? Explain why

CASE-3 [3 marks]: **"A mobile learning application collects student names, locations, and usage patterns. Privacy settings are added only after parents complain".** What privacy principle is violated?

CASE-4 [3 marks]: **"A smart parking application collects drivers' location data to find nearby parking spaces. Later, the company shares this location history with marketing agencies".** Identify the privacy threat and suggest a privacy-respecting alternative.

CASE-5 [3 marks]: **"An online shopping website wants to protect customer accounts from phishing emails that steal passwords".** Recommend a stronger authentication method and explain how it improves security.

P3-Q2 [10 marks]: A company information system manages employee resources. The system has the following subjects and objects. Construct the Access Control Matrix for this system.

Subjects (Users):
- HR Manager,
- Project Leader
- Employee

Objects (Resources):
- Employee Records
- Project Documents
- System Logs

Access Rights:
- Read (R)
- Write (W)
- Execute (X)
- Delete (D)

The security policy specifies:
- HR Manager has full access to Employee Records and read-only access to Project Documents.
- Project Leader has Read/Write access to Project Documents only.
- Employee has Read-only access to Project Documents only.
- Only the HR Manager can delete Employee Records.
- Only the HR Manager can access Employee Records.
- No user except the HR Manager can access System Logs, and access to System Logs is Read-only.

End of questions

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